

Final Project

Phase - I Planning

Raja Ranjith Kumar Asileti (1228369443)

Hero-Godsway Zilevu

Chibuikem Umeh

Arizona State University – Poly Campus

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Professor: Asmaa Elbadrawy

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Section 1: Dataset Description

This expansive dataset encapsulates 120 years of Olympic history, spanning from 1896 to 2016, offering a detailed exploration of athletic events and their outcomes. The dataset not only provides insights into the physical attributes of athletes, such as their height, weight, and medal achievements, but also delves into the diverse array of sports that have graced the Olympic stage. From the exhilarating pace of hockey and the strategic finesse of basketball to the agility of badminton and beyond, each sport contributes to the vibrant tapestry of Olympic history.

In addition to individual accomplishments, the dataset unravels the narrative of Olympic events, detailing the locations, years, and seasons that have witnessed the pursuit of athletic excellence. Moreover, it sheds light on the myriad sports embraced by the Olympics, unveiling the rich diversity of competitions, from track and field to team sports like hockey and basketball. This comprehensive dataset invites exploration into the triumphs, challenges, and enduring spirit of athleticism, offering a holistic view of the Olympic Games throughout the years.

Rows: 217K

Columns : 15

This section explains the characteristic attributes of the data gathered.

Columns	Data Type Description	Domain & Possible Use Case
ID	This column contains individual identifiable participant ID numbers for all athletes involved in the Olympics game. It is a nominal data	The ID column can be used as identifiers similar to the name which can help track an athlete of interest in multiple visualizations
Name	This column provides participant names. Data type is nominal	The name column can help to find the specific athlete

Sex	This column provides categorical data that groups participants by biological sex	The sex column provides information that can be used to compare different genders and takes values of Male or Female.
Age	This column provides information about the age of the participants. The data type is ratio	This column provides information that enables comparison between different age groups. It contains integers between 9 - 100 measured in years
Height	This column provides information on the height of participating athletes. The data class is ratio	Height is a ratio attribute measured in centimeters. This values could be analyzed to understand how physical attributes influence performance
Weight	This column contains information on the weight of players.	Weight is measured in kilograms and could be combined with other attributes to understand how it may influence performance
Team	Team is a nominal attribute that contains information about the country athlete is representing	This could also be used to compare different country performances in the Olympics
NOC	NOC is the code used to represent the country of an athlete. The data type is nominal	This information could be used for axis labels, or as prettentive attributes in the visualization
Games	It is a nominal data which contains information the describes the year and season of the olympics tournament the athlete participated in	This column combines information between different years and seasons.
Year	This is an ordinal data that shows the year in which the event happened	This is an ordinal attribute that shows calendar years
Season	The season is a categorical	Season contains categorical

	dataset that tells when the event took place	labels of Winter or Summer. It could be used to compare how performance records of different teams in different seasons.
City	This column contains nominal data about the hosting city of the event	Nominal label of describing city in which the event took place.
Sport	This column contains categorical attributes of the type of sport the athlete participated in.	These attribute denotes the sport name such as curling, Ice hokey etc,. in text format.
Event	Contains categorical information describing the specific type of sport	This column could be used as filter to allow users to see information about name of the sport event they are participating
Medal	Ordinal Data type which contains information that describes the outcome from the game participation.	This column takes the values of gold, silver, bronze or NA(Not Achieved).

Section 2: Prospective Dashboard Users

Athletes: Athletes can utilize the dashboard to benchmark their performances against historical data, gaining insights into successful strategies and areas for improvement. They can analyze the performances of past Olympic medalists, understand the competitive landscape, and make informed decisions about training priorities. The dashboard provides a valuable resource for athletes to set realistic goals, track progress, and draw inspiration from the achievements of their predecessors. Additionally, athletes can use the dashboard to explore the history of their respective sports, learn about the experiences of iconic athletes, and connect with the rich heritage of the Olympic Games.

Athlete Coaches: Coaches can leverage athlete-specific information such as height, weight, and performance metrics to tailor training programs. They can identify successful training regimens, evaluate the impact of specific physical attributes, and make data-driven decisions to enhance athletes' preparation for Olympic competitions.

Sports Analysts: Sports analysts can delve into the dashboard to perform statistical analyses, identify performance trends, and generate insights into the strategies employed by athletes and teams over the course of Olympic history. They can examine medal counts, winning patterns, and correlations between athletes' physical attributes and success.

Event Planners: Event planners can use historical data on Olympic events, including venue choices, logistical details, and audience engagement, to optimize the planning and execution of future Olympic Games. Insights from past events can inform decisions on venue selection, scheduling, and infrastructure development.

Section 3: List of User Requirements

User	Requirements
Athletes	As prospective olympic participants, athletes may use this dashboard to track how physical parameters such as weight, age affect their chances of winning their respective competition. They may also be interested in seeing how many competitors that they have to compete against
Athlete Coaches	Coaches may use this dashboard to set the physical standards that athletes must exceed in order to be considered to compete. The coaches could also use this dashboard to analyze patterns in previous performances that may help improve their chances of winning.
Sports Analysts	Sports analysts could use the dashboard to adequately compare how physical fitness affects the chances of winning. They could use this dashboard to estimate who the favorites for each competition could be based on their historical performance.
Event Planners	Event planners may be interested in seeing the number of participants per each competition type, and how those have changed over time to see how to prepare for the subsequent events.

Potential Questions

1. How many Athletes participated in the respective event based on season(Summer or Winter)
2. Which Country Won Gold, Silver, Bronze in these 120 years of Olympic events.
3. In Which City the olympic events were conducted got more Gold, silver and bronze medals.
4. Has there been an increase in female participation over the years particularly in often male dominated sports

5. What is the distribution of the age of gold medalists in either the summer or winter seasons?
6. Show globally the number of acquired medals, has there been a fair spread/distribution?
7. In which Olympic sport or discipline has the United States achieved the fewest gold medals over the course of the 120-year history of the Games?
8. What is the variations for female participation for the different seasons over the years
9. For selected teams(countries), what is the average age of women and men participating in the Olympics over the years
10. For the past years, what is the average BMI of medal winners for different competitions over the years?

References

Dataset Link: Depinto, M. G. (2018, September 23). Let's discover more about the Olympic Games! Kaggle.

<https://www.kaggle.com/code/marcogdepinto/let-s-discover-more-about-the-olympic-games/notebook#7.-Medals-per-country>

Mural 23 Link:

<https://app.mural.co/t/datavisualizationfinalprojee5159/m/datavisualizationfinalprojee5159/1699563637854/1cc71ea8316fec9084fe918b1eabef52bdd8549e?sender=uc73c99f02de719af19555441>